

Prediction Inputs

Select Junction

1

▼

Select Date

2026/06/22

Select Hour

12



Smart City Traffic Forecasting Dashboard

Smart City Traffic Management System

This dashboard predicts traffic volume at city junctions using a Random Forest Machine Learning model. The system helps authorities anticipate congestion and support infrastructure planning.



Model Performance

MAE

2.39

RMSE

3.54

R² Score

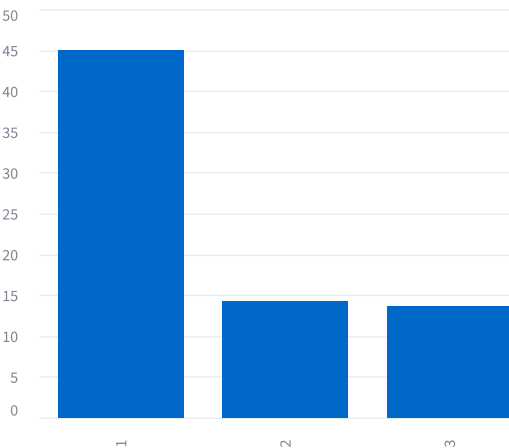
0.969

 Predict Traffic

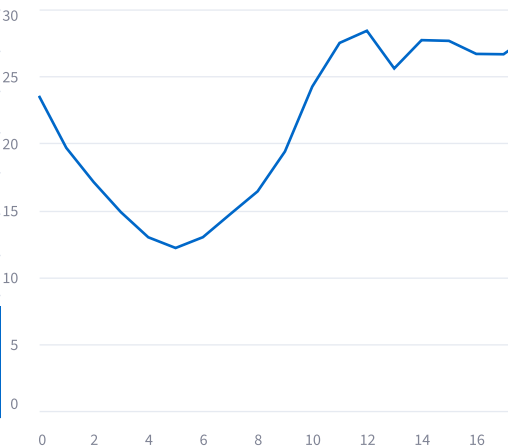


Traffic Analysis

Average Traffic by Junction



Average Traffic by Hour



Monthly Traffic Trend



Weekend vs Weekday Traffic



Project Information

Project Title: Forecasting of Smart City Traffic Patterns

Objective: Forecast traffic patterns across four city junctions to help city authorities manage congestion and improve infrastructure planning.

Machine Learning Model: Random Forest Regressor

Features Used:

- Junction
- Year
- Month
- Day
- Hour
- Day Of Week
- Weekend Indicator
- Holiday Indicator (IsHoliday)

Key Outcomes:

- Identified traffic peaks.
- Analyzed traffic trends across four junctions.
- Incorporated holiday effects.
- Generated future traffic forecasts.
- Built an interactive dashboard for decision-making.



Internship Project Completed Successfully